

Shubham Mishra

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EDUCATION

Netaji Subhas University of Technology

Bachelor of Technology (B.Tech)

Delhi, India

2022 – 2026

EXPERIENCE

Embedded Systems Intern

Emertxe Technologies

May 2025 – July 2025

Remote

- Developed and simulated a PIC microcontroller-based embedded system using Embedded C, MPLAB IDE, and PICSIMLAB, validating functionality across multiple operational states.
- Implemented state-machine logic, timers, CLCD interfaces, and actuator control workflows while performing debugging, testing, and real-time system validation.

PROJECTS

Vehicle Insurance MLOps Pipeline | *Python, Scikit-learn, MongoDB, AWS, Docker*

(GitHub)

- Built an end-to-end machine learning pipeline for vehicle insurance prediction using Python, MongoDB, Pandas, NumPy, and Scikit-learn, implementing data ingestion, validation, transformation, feature engineering, model training, and model evaluation workflows.
- Designed ETL pipelines for efficient data processing, data validation, and automated retraining while ensuring reproducibility and scalability across the ML lifecycle.
- Implemented MLOps practices using AWS, Docker, GitHub Actions, and CI/CD pipelines to automate model deployment, monitoring, and production workflows.

Pneumonia Detection MLOps Pipeline | *PyTorch, MLflow, DVC, Docker, AWS, Flask*

(GitHub)

- Developed a deep learning solution for pneumonia detection from chest X-ray images using PyTorch, implementing data preprocessing, augmentation, model training, validation, and performance evaluation.
- Established reproducible machine learning workflows using MLflow for experiment tracking and DVC for data and model versioning, enabling systematic comparison of model performance.
- Containerized deployment workflows using Docker and automated CI/CD pipelines with GitHub Actions, AWS ECR, and EC2 for scalable model serving through a Flask inference API.

AI Travel Agent Chatbot | *Python, LangChain, LangGraph, Redis, HuggingFace, Ollama*

(GitHub)

- Developed an intelligent AI travel assistant using LangGraph and Redis to deliver personalized recommendations through context-aware reasoning and memory management.
- Implemented semantic vector search, memory retrieval, smart deduplication, and LLM-powered tool execution for efficient information retrieval and long-term user personalization.
- Designed scalable workflow orchestration pipelines for user interaction, context management, decision-making, and personalized recommendation generation.

AI Water Tracker App | *Python, FastAPI, Streamlit, SQLite, Ollama*

(GitHub)

- Built a full-stack AI-powered hydration tracking platform using FastAPI, Streamlit, SQL, and Ollama.
- Developed RESTful APIs and data processing pipelines for data ingestion, validation, storage, aggregation, and retrieval of user health metrics.
- Integrated LLM-based analytics to generate personalized hydration recommendations using structured data insights and automated reasoning workflows.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, HTML/CSS

ML/AI: Scikit-learn, PyTorch, TensorFlow, NumPy, Pandas, LangChain, LangGraph, OpenAI API, Feature Engineering, Model Evaluation

Data & MLOps: ETL Pipelines, Data Ingestion, Data Processing, Docker, MLflow, DVC, CI/CD, GitHub Actions

Cloud & Frameworks: AWS (EC2, ECR), FastAPI, Flask, Streamlit, REST APIs

Databases & Tools: MongoDB, PostgreSQL, SQLite, ChromaDB, FAISS, Qdrant, Git, Linux, VS Code